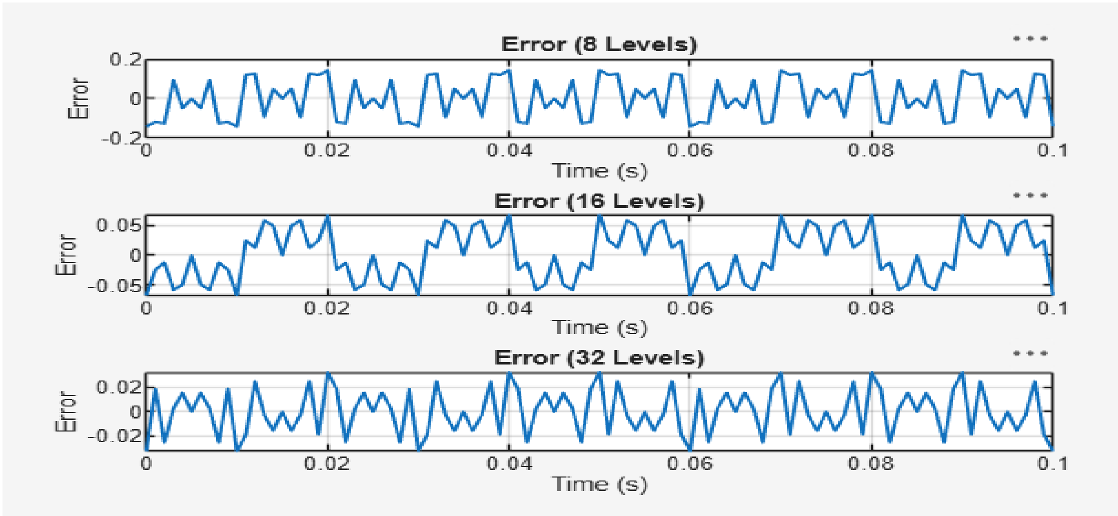
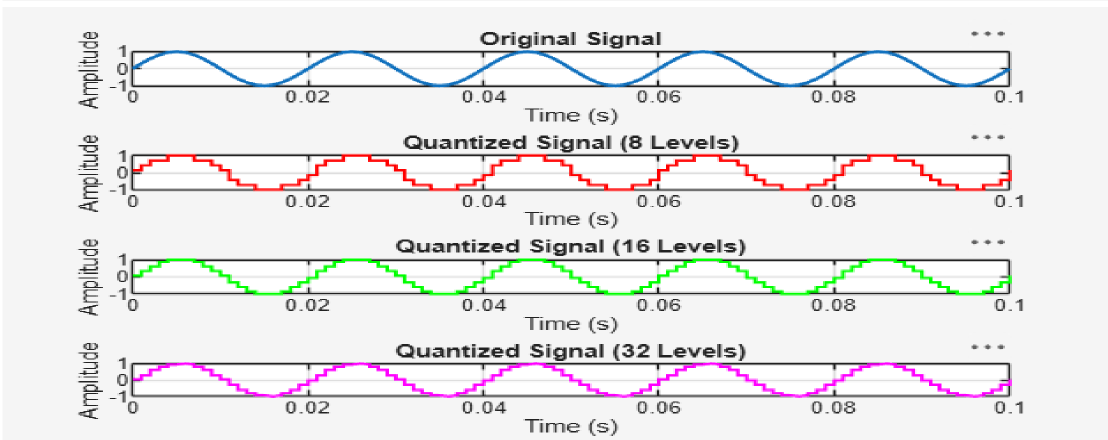
Experiment 6

6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal

Code:

```
clc; clear; close all; %% Signal fs = 1000; f = 50; t = 0:1/fs:0.1; x = sin(2*pi*f*t); %% Quantization Levels L = [8 16 32]; colors = {'r','g','m'}; figure; for i = 1:length(L) % Quantization xq{i} = round((x+1)(L(i)-1)/2)(2/(L(i)-1)) - 1; % Error & MSE e{i} = x - xq{i}; mse(i) = mean(e{i}.^2); % Plot Quantized Signals subplot(4,1,i+1); stairs(t,xq{i},colors{i},'LineWidth',1.2); title(['Quantized Signal (' num2str(L(i)) ' Levels)']); grid on; end % Original Signal subplot(4,1,1); plot(t,x,'LineWidth',1.5); title('Original Signal'); grid on; % Display MSE fprintf('MSE for 8 Levels = %f\n', mse(1)); fprintf('MSE for 16 Levels = %f\n', mse(2)); fprintf('MSE for 32 Levels = %f\n', mse(3)); %% Error Plots figure; for i = 1:3 subplot(3,1,i); plot(t,e{i}); title(['Error (' num2str(L(i)) ' Levels)']); grid on; end
```





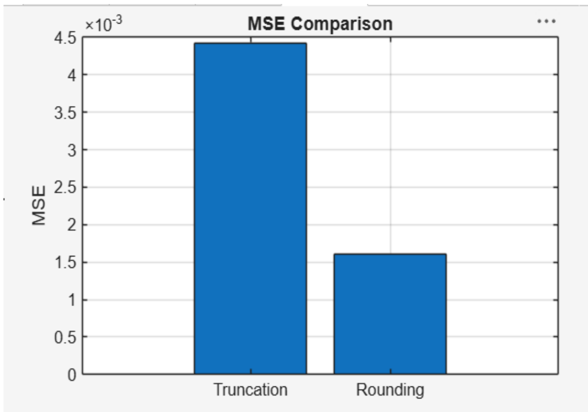

```
MSE for 8 Levels = 0.010473
MSE for 16 Levels = 0.001763
MSE for 32 Levels = 0.000357
```

6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation Rounding Techniques

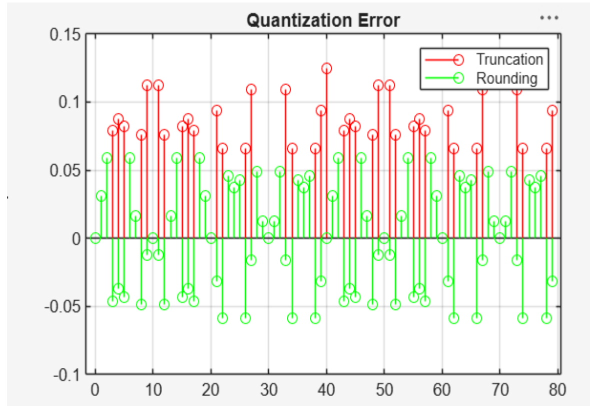
Code:

```
clc; clear; close all n=0:79; x=sin(2*pi*n/40); q=8; xt=floor(x*q)/q; xr=round(x*q)/q; et=x-xt;
er=x-xr; % Fig 1 figure plot(n,x,'b',n,xr,'r','LineWidth',1.5) grid on title('Original vs Quantized')
legend('Original','Quantized') % Fig 2 figure plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5) grid on
title('Truncation vs Rounding') legend('Original','Truncated','Rounded') % Fig 3 figure
stem(n,et,'r') hold on stem(n,er,'g') grid on title('Quantization Error')
legend('Truncation','Rounding') % Fig 4 figure bar([mean(et.^2) mean(er.^2)])
set(gca,'XTickLabel',{'Truncation','Rounding'}) grid on title('MSE Comparison') ylabel('MSE')
```

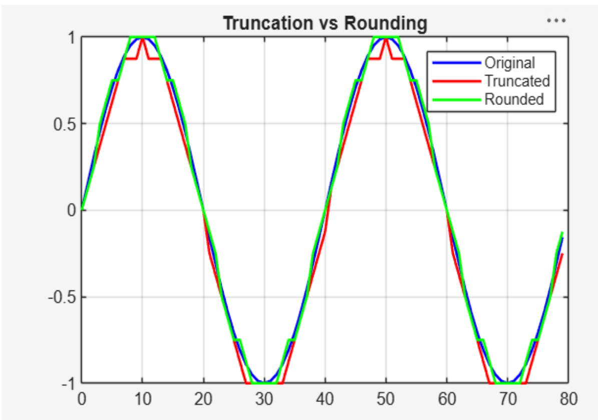
Output:



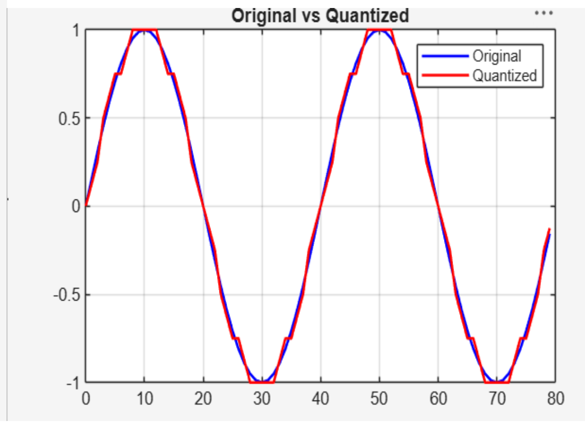
MSE Comparison
×10-3
MSE
Truncation
Rounding



Quantization Error
Truncation
Rounding



Truncation vs Rounding
Original
Truncated
Rounded



Original vs Quantized
Original
Quantized